Type MS · Input, process, output and other computational models

Answer 1 of 7 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part b.ii in this mark scheme extract

2(a)	A1	**The only correct answer is D** ***A*** *is not correct because abstraction is a technique used in computational thinking* ***B*** *is not correct because encryption is a security measure* ***C*** *is not correct because iteration is a programming construct*		**1**

Question	mp	Answer	Additional guidance	Mark
2(b)(i)		Award **one** mark for each correct cell:		
	B1 B2	real/double/float: (1) • pay rate/18.75 (1) • total pay/356.25 (1) • tax/0.2/71.25 (1) • income/285.00 (1) int/integer: (1) • hours/19 (1)	Example must match data type for B2 Award suitable example even if not taken from payslip Accept string and matching example e.g. Hours	**2**

5 of 23

Type MS · Input, process, output and other computational models

Answer 1 of 7 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part b.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
2(b)(ii)		Award **one** mark for each correct cell:		
	B3	**Input:** • Pay Rate/18.75 (1) • Hours/19 (1)		
	B4	**Process**: • Pay Rate * Hours/18.75 * 19 (1) • Total Pay * Tax/356.25 * 0.2 (1) • Total Pay – Tax/356.25 – 71.25 (1)	Don't need output e.g. total pay, just the calculation part e.g. pay rate * hours: Total pay = pay rate * hours Accept 0.2 or 20% as part of calculation for tax Ok to mix types e.g. Total Pay * 20% Accept descriptions such as *multiply the pay rate by hours (hours etc could be taken from input box/ B3 mark)*	
	B5	**Output:** • Total Pay/356.25 (1) • Tax/71.25 (1) • Income/285 (1)		**3**

6 of 23

Type MS · Input, process, output and other computational models

Answer 1 of 7 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part b.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
2(c)(i)	C1 C2	Award **one** mark for **one** correct cell: Award **two** marks for all **three** correct cells: • **Normal:** any value between 0-255 e.g. 125 • **Boundary:** 0/255 • **Erroneous:** any negative value/any value 256 or above	Accept -1/256 for boundary data Don't accept non-int types	**2**

Question	mp	Answer	Additional Guidance	Mark
2(c)(ii)		Award **one** mark for any of the following up to a maximum of **four** marks:		
	C3	Input changed to int before comparison (1)		**4**
	C4	Relational operator used correctly to check for valid input (both **>=**0 and **<**256 or equivalent) (1)	Allow: >0 and/or <255	
	C5	'and' logical operator used to combine relational tests (1)	or operator acceptable if candidate rearranges the selection statement and test works for correct code option – check if C4 is now awardable	
	C6	Calling of `getHex(userInput)` (1)	Call must be inside print ( ) Subprogram and argument must be functionally correct Ignore additional variables/strings in print statement	

7 of 23

Type MS · Input, process, output and other computational models

Answer 1 of 7 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part b.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
37 # complete the line to convert the user input to integer
38 userInput = int(input("please enter a number between 0 and 255 \n"))
39
40 # complete the test to check the input is between 0 and 255
41 if (userInput >=0) and (userInput <256):
42     # complete the line to call getHex subprogram
43     print(getHex(userInput))
44 else:
45     print("number out of range")
```

8 of 23

Type MS · Input, process, output and other computational models

Answer 1 of 7 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part b.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Input, process, output and other computational models

Answer 2 of 7 · Question 1 part a · 3 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

1(a)	Award **one** from: Input, process, output in this order – **3 marks** Input, process, output in any order – **2 marks** At least one of input, process, output – **1 mark**		3

Question Number	Answer	Additional Guidance	Mark
1(b)	B: Central processing unit **The only correct answer is B** **A** is not correct as this is memory only **C** is not correct as this is memory only **D** is not correct as this is used only for transporting memory addresses		1

Question Number	Answer	Additional Guidance	Mark
1(c)	• Data (1) • Instructions (1)		2

Question Number	Answer	Additional Guidance	Mark
1(d)	Any **two** from: 1. Virtual memory (VM) is used when RAM becomes full (1) (to hold all programs and data). 2. Virtual memory is used as (an extension to) main memory/RAM / works like RAM. (1) 3. Virtual memory is stored/created on (internal) secondary storage/HDD/SSD. (1) 4. Virtual memory is used as temporary storage. (1) 5. Instructions and data not currently being used are transferred from RAM to VM/HDD. (1) 6. When needed again, instructions and data are transferred back to RAM. (1)		2

Type MS · Input, process, output and other computational models

Answer 2 of 7 · Question 1 part a · 3 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Any **two** from: 1. The disc surface is made out of polycarbonate. (1) 2. A laser (beam) is used. (1) 3. Data is stored as a series of tiny grooves arranged in tracks on the surface of the disc. (1) 4. Bumps/lands/flats/troughs/pits/dips imprinted/burnt on the disc. (1) 5. Bumps and flats (etc.) represent 1 and 0 (bits) / the reflection (of the laser) represents 1 and 0 (bits). (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)	D: System software **The only correct answer is D** **A** is not correct because this category is purely for software that is used for security purposes **B** is not correct because this category is designed for end user tasks rather than for managing the computer itself **C** is not correct because this software is used to translate a high level program into machine code		1

Type MS · Input, process, output and other computational models

Answer 2 of 7 · Question 1 part a · 3 marks

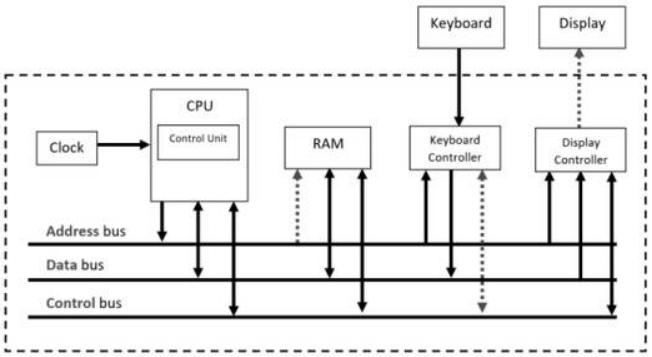
2019 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Input, process, output and other computational models

Answer 3 of 7 · Question 3 part c · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

3(a)	 • **Address bus** labelled correctly (1) • **Data bus** labelled correctly (1) • **Control bus** labelled correctly (1) • Arrow from address bus to RAM (1) • Bi-Directional arrows from control bus to keyboard controller (1) • Arrow points from display controller to display (1)		6

Question Number	Answer	Additional Guidance	Mark
3(b)	Award **two** marks for a linked explanation such as: The cache stores frequently used data/instructions (1) so the processor does not have to wait because cache is checked before main memory / because RAM is further away from the processor (1). It speeds up processing (1) because cache is a faster type of memory (1).		2

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Type MS · Input, process, output and other computational models

Answer 3 of 7 · Question 3 part c · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	<table><tr><th>CPU1</th><th>CPU2</th></tr><tr><td>5 * 2 = 10</td><td>4 * 3 = 12 (1)</td></tr><tr><td>10 + 12 = 22 (1)</td><td>24 / 4 = 6 (1)</td></tr><tr><td>22 - 6 = 16 (1)</td><td></td></tr></table>	The second row expressions can be in either column 22 – 6 = 16 May appear in either CPU in the final row	4

Question Number	Answer	Additional Guidance	Mark
3(d)	Any four from: • The OS checks whether sufficient space is available on the storage (media) for the file (1). • The file is broken into blocks (1) • The blocks are stored in spaces that are large enough (1) • OS looks for (fat/ntfs)/ reserves (linux) (groups of) contiguous blocks (1) • Blocks can reside anywhere on the storage (1). • The OS updates the file allocation table (with the start location and sequence number of each block) (1) • OS sets hard links to files (1) ntfs • OS updates journal/MFT (1) ntfs/linux • Metadata about the file such as read/write permissions, date created and last accessed is separately stored (1)		4

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Type MS · Input, process, output and other computational models

Answer 3 of 7 · Question 3 part c · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Input, process, output and other computational models

Answer 4 of 7 · Question 1 part d · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · Input, process, output and other computational models

Answer 4 of 7 · Question 1 part d · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

Type MS · Input, process, output and other computational models

Answer 4 of 7 · Question 1 part d · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(e)(i)</td><td>Award one mark from:

• When fast execution of the program is essential/gives faster execution (1)
• When writing code to control a device / directly access the hardware (1)
• To make efficient use of a device's limited storage/memory/power (1)
• When writing code for a particular architecture/hardware/embedded system (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(ii)</td><td>Award one mark for:

• To translate/convert assembly language into machine/object code (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(iii)</td><td>Award one mark for any two correct rows (1)
Award two marks for any four correct rows (2)
Award three marks for all five correct rows (3)

<table><tr><th>Description</th><th>Compiler</th><th>Interpreter</th></tr><tr><td>Translates the program each time it is executed</td><td></td><td>☐</td></tr><tr><td>Produces permanent object code</td><td>☐</td><td></td></tr><tr><td>Translates line by line</td><td></td><td>☐</td></tr><tr><td>Translates the whole program before it is run</td><td>☐</td><td></td></tr><tr><td>Generates a list of errors once the complete program has been translated</td><td>☐</td><td></td></tr></table></td><td>Row can only be correct if there is only one tick.</td><td>3</td></tr>
<tr><td colspan="3">Total for question 1</td><td>11</td></tr>
</table>

Type MS · Input, process, output and other computational models

Answer 4 of 7 · Question 1 part d · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part a in this mark scheme extract

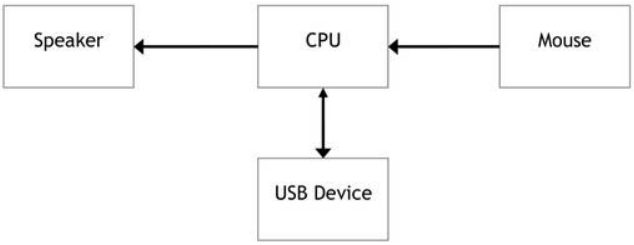
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part a in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part a in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · Input, process, output and other computational models

Answer 5 of 7 · Question 5 part a · 2 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Input, process, output and other computational models

Answer 6 of 7 · Question 1 part a · 3 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Input, process, output and other computational models

Answer 6 of 7 · Question 1 part a · 3 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Input, process, output and other computational models

Answer 6 of 7 · Question 1 part a · 3 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Input, process, output and other computational models

Answer 7 of 7 · Question 4 part b · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

4(a)	Award up to **two** marks for a linked explanation such as: • 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1) • The simulation can be rerun with different values (1) to see how the outcomes will change (1) • The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1) • The simulation can be safer (1) because of the danger of animal interaction (1) • The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1)	Looking for a relevant statement and a linked extension/expansion. Do not award two independent statements that are not linked. Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ.	2
4(b)	**The only correct answer is C** ***A** is not correct because it is not a linear scenario* ***B** is not correct because parallel models do not have to communicate and coordinate their actions* ***D** is not correct because there is not an explicit input with an expected output*		1

Type MS · Input, process, output and other computational models

Answer 7 of 7 · Question 4 part b · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

4(c)	Award up to **two** marks for a linked description such as: • The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1) • Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1) • The research facility could implement a security question (1) that only the employee would know the answer to (1)	Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question. Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated	2
4(d)(i)	Award **one** mark for any of the following: • The manipulation of matter with a size from 1 to 100 nanometres / scale of 10^{-9} / at atomic/molecular level (1) • The inclusion of nanoparticles into different materials (1) • Facilitates miniaturisation at atomic/molecular **scale/level** (1)	Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level.	1
4(d)(ii)	Award **one** mark for any of the following: • Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1) • Miniaturisation of electronics e.g. transistors (1) • Nanomedicines to deliver drugs (1)	Allow examples that reasonably map to the categories given.	1

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Type MS · Input, process, output and other computational models

Answer 7 of 7 · Question 4 part b · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

4(e)	Award **one** mark for each of the following up to a maximum of **three** marks: • It uses a total of 128 bits (1) • There is a total of 32 hexadecimal character (1) • Each hexadecimal character requires 4 bits (1) • There are 8 blocks of 4 character / 8 x 4 blocks (1) • Each block (of 4 character) requires 16 bits (1) • The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1) • Each of the 8 blocks uses 16 bits / 8 x 16 (1)		3
4(f)	Indicative content • Rights and responsibilities • Data Protection regarding personal data • Surveillance • Consent of the individual to having specified data held • Accountability of those running / using such systems • Ownership • Data mining & analysis of work patterns and potential use/misuse • Ethics versus the legality of implementing such systems • Data misuse and protection against hacking and malware • Requirement to keep data secure • Physical access to areas of the site • Monitoring employee movement to restricted areas / resources • Algorithmic bias in facial recognition		6
		Total for question 4	**16**

Type MS · Input, process, output and other computational models

Answer 7 of 7 · Question 4 part b · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 7 of 7 · Question 4 part b · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark